



THE UNIVERSITY OF
NOTRE DAME
A U S T R A L I A

MED 4000

WEEK 21

TUTOR GUIDE

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The cases and materials in this guide were reviewed by members of the University of Notre Dame Clinical Years Curriculum Committee (CYC).

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RELEVANT PRIOR LEARNING

MED 1000

- 1.31 Anatomy of the orbit & eye. [L]
- 1.31 Physiology of vision. [L]

MED 2000

- 2.03 Applied Anatomy of the orbit & eye. [L]
- 2.03 History and examination of the eye. [L & CCS]
- 2.03 Eye trauma & minor eye surgery [L]

LEARNING OBJECTIVES:

WEEK 21 _SURGERY _OPHTHALMOLOGY		
Yr	Wk	LEARNING OBJECTIVES:
4	21	• List and briefly describe the typical patterns of traumatic injury to the eye, with reference to the anatomical and • Know the hobbies or occupations that pose a particular risk for penetrating eye injuries. • Demonstrate a safe approach to the clinical assessment of ocular trauma or chemical injuries. • Describe and demonstrate the basic emergency measures required when treating ocular trauma or chemical injuries. • Know the indications, in the setting of ocular trauma, for urgent referral to an ophthalmologist.
4	21	• Describe the historical and examination findings that assist in differentiating between viral, bacterial and allergic conjunctivitis. • List the common pathogens involved in bacterial ocular infections. • Understand the non-drug treatment of various types of ocular infections. • Know the most appropriate empirical and directed anti-microbial therapies for common ocular pathogens. • Describe the classical presentation of Herpes virus infections and outline measures that should be taken to ensure this condition is not overlooked. • Describe the clinical features of keratitis, pre-septal cellulitis and peri-orbital cellulitis. • List the complications that may arise from serious or poorly treated ocular infections.
4	21	• List the differential diagnosis of the unilateral red eye and be able to identify which of these are potentially sight threatening. • Demonstrate a safe approach to the assessment of the unilateral red eye. • Describe the assessment and treatment of uveitis, ocular foreign body, acute glaucoma and viral keratitis. • List the systemic diseases that commonly present with ocular manifestations. • Briefly describe the common types of inflammatory conditions of the eye. • Differentiate between uveitis, scleritis, episcleritis and understand the basic elements of treatment. • Know the indications for urgent specialist referral of a unilateral red eye.

WEEK 21_SURGERY_OPTHALMOLOGY		
Yr	Wk	LEARNING OBJECTIVES:
4	21	• Briefly describe the basic on-site first aid measures for chemical exposures to the eye. • Discuss the factors that are often associated with a poor outcome in chemical injuries to the eye. • Discuss the sequence of assessment and management that is recommended in the emergency department setting following ocular chemical exposure. • Identify the criteria for deciding when it is safe to cease irrigation of the eye. • Discuss the criteria for determining the severity of a chemical eye injury. • Identify the indications for the urgent involvement of an ophthalmologist.
4	21	• For patients with sudden visual loss, explain the different presenting features of: a) pathology involving the optic tract or cerebral cortex, versus b) pathology anterior to the optic chiasm. • List the common causes of sudden visual loss involving one eye. • Identify the important differentiating features on history between the various causes of sudden visual loss, with a particular emphasis on time sequence, associated symptoms and co-morbidities. • Identify the most important signs to elicit in the patient with sudden visual loss involving one eye. • Outline the common investigations that might assist in determining the cause or assisting the treatment of patients with sudden visual loss in one eye.

CASE ONE

Short case number: 4_21_01

Category: Nervous System and Eye

Discipline: Surgery

Setting: Emergency Department

Topic: Eye trauma and injury

CASE



Aaron Balcombe is 17 years old and works as an apprentice motor vehicle mechanic with his uncle Paul's small business.

Aaron has presented to the emergency department of his local hospital after being struck in the eye by a spanner whilst working on a vehicle.

"We need some help here!" shouts Paul as he helps Aaron through the sliding doors of the department. Aaron's face is buried in an ice pack that has been covered with a tea towel. The tea towel is tinged with red.

QUESTIONS

1. What information should be sought from Aaron and his uncle?
2. In general, what are the key components of examination, in assessing ocular trauma?
3. What are the common patterns of non-penetrating (blunt) injury to the eye?
4. What are the common patterns of penetrating injury to the eye?
5. Aside from definitive operating theatre procedures, briefly outline the important principles in the emergency management of each of the following:
 - Hyphaema
 - corneal abrasion and foreign body
 - ruptured globe
 - intra-ocular foreign body
 - retinal tears/haemorrhages/detachment
 - orbital floor fracture

Resources

- James B, Chew C, Bron AJ. Lecture notes on ophthalmology. Wiley-Blackwell; 2003.
- Rudkin AK, Lake S. How to treat: Eye Injuries. Australian Doctor [Internet]. 2009 May 22; Available from: http://www.australiandoctor.com.au/htt/pdf/AD_027_034_MAY22_09.pdf
- Khaw PT, Shah P, Elkington AR. Injury to the eye. BMJ. 2004;328(7430):36.

ANSWERS

1. What information should be sought from Aaron and his uncle?

- Mechanism of the injury
High vs. low velocity
Time and place.
- Context (wearing glasses/goggle?)
- Current symptoms.

2. In general, what are the key components of examination, in assessing ocular trauma?

WITH the naked eye, inspect the face, peri-orbital region and eyelids, noting any bruising, swelling, lacerations, foreign bodies or asymmetry. Palpate the orbit, noting areas of tenderness or hypo-aesthesia. In the setting of trauma, the most important tests of visual function are acuity and pupil response. Other tests include assessment of visual fields, brightness and colour.

Visual acuity

Ascertaining visual acuity at the first assessment is essential: it is a measure of visual function generally, but it also has prognostic significance in some injuries.

If visual acuity is less than 6/6, use a pinhole to determine whether vision can be improved. If it is not possible to use the Snellen chart at six metres, a near reading chart may be substituted.

Pupil response

Unlike other tests of visual function, pupil response is an objective measure. The presence of an afferent pupillary defect indicates optic nerve or severe vitreo-retinal injury. It is not caused by injuries in the anterior segment of the eye. The presence of a relative afferent pupillary defect (RAPD) after a traumatic injury is an ominous sign and requires urgent investigation.

Testing eye movements

Determine whether the eyes are straight in primary gaze, then assess eye movements by having the patient look in all directions. While testing, ask the patient if they experience double vision. Are the images horizontal, vertical or oblique to each other? Does one image disappear if an eye is closed? Binocular diplopia is consistent with restricted motility; monocular diplopia may be secondary to a dislocated lens or iris root tear.

In the context of trauma the potential causes of strabismus are varied. Mechanical restriction secondary to muscle entrapment or generalised swelling is common, although paralytic lesions must also be considered.

Testing eyelids

Ptosis in the context of eyelid trauma might be the result of lid swelling, although other causes might be considered, such as injury to the eyelid tendon (the levator aponeurosis) or to the levator itself. Ptosis without an adequate explanation requires prompt evaluation to exclude sympathetic (Horner's) or third nerve palsy.

Examine the anterior segment

Examine the eye from front to back using whatever equipment is available, whether this is a pen torch or slit lamp.

Lids and conjunctiva

Examine the external lids for a laceration, noting particularly whether it involves the lid margin or lacrimal drainage apparatus. If there is no obvious open injury to the eyeball, evert the lids. Even small subtarsal foreign bodies can cause severe inflammation and large corneal abrasions, so examination needs to be thorough.

Cornea

If a slit lamp is not available use a pen torch to examine the cornea from a variety of angles. Ask the patient to look in all directions to view the entire cornea. Look carefully for foreign bodies. After this, stain the eye with fluorescein and examine with a cobalt blue light. Areas of denuded epithelium will fluoresce under blue light. A corneal abrasion should prompt a thorough search for a foreign body, and both lids need to be everted.

Sclera

Examination of the sclera can be difficult, as even the most trivial trauma can cause a subconjunctival haemorrhage that will obscure your view. Look carefully for wound leakage and frank lacerations, and for patches of dark pigmentation, which may indicate exposed choroid (an inner layer of the retina).

Anterior chamber

Inspect the anterior chamber carefully, particularly for blood (hyphaema). Compare the anterior chamber of each eye: they should be symmetrical. A very deep or very shallow anterior chamber in the context of trauma may indicate an open eye injury.

Iris and pupil

Aside from testing pupillary response, note the contour of the pupil. Peaked irregular pupils often indicate a penetrating injury.

Lens

Dislocation or opacities of the lens can be seen when assessing the red reflex with the direct ophthalmoscope. Examine the posterior segment (vitreous and retina) In the first assessment of a patient with trauma, the role of fundoscopy with a direct ophthalmoscope is limited. Because it provides a limited view of the fundus, an unremarkable fundus examination cannot be considered reassuring.

The sudden onset of photopsia (flashing lights), floaters, a drop in acuity not adequately explained by anterior segment pathology, or visual field loss are more important in triaging a patient than the findings on direct fundoscopy. The presence of any of these symptoms should prompt immediate referral to an ophthalmologist for a dilated fundus examination.

3. What are the common patterns of non-penetrating (blunt) injury to the eye?

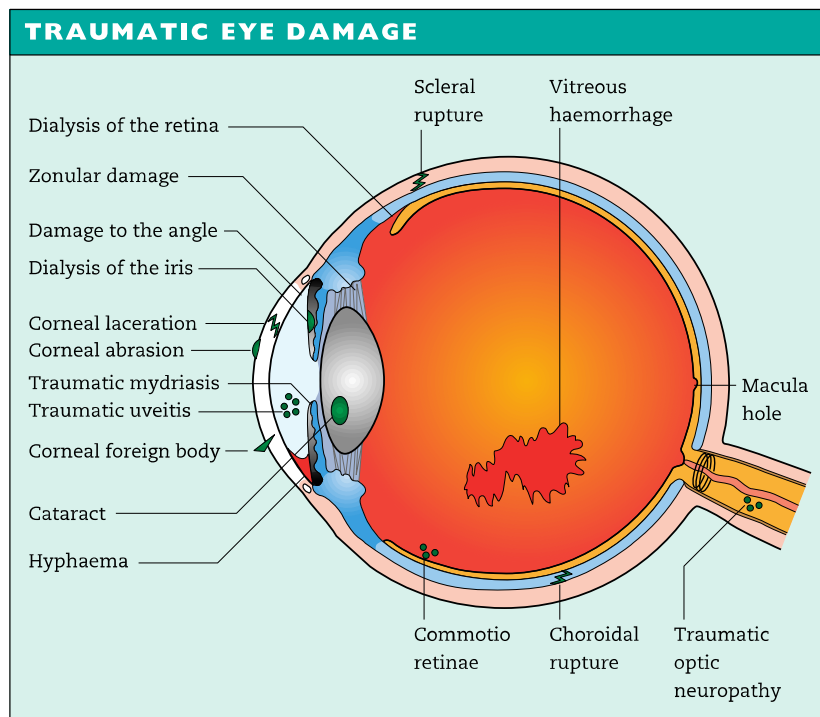


Fig. 16.1 The extent of possible traumatic damage to the eye.

When a large object (such as a football) hits the eye, most of the impact is usually taken by the orbital margin. However, if a smaller object (such as a squash ball) hits the area, the eye itself may take most of the impact. Haemorrhage may occur and a collection of blood may be visible in the anterior chamber of the eye (hyphaema). Patients who sustain such injuries need to be reviewed at an eye unit as the pressure in the eye may rise, and further haemorrhages may require surgical intervention.

Haemorrhage may also occur into the vitreous or in the retina, and this can be accompanied by a retinal detachment. All patients with visual impairment after blunt injury should be seen in an ophthalmic department.

The iris may also be damaged and the pupil may react poorly to light. This is particularly important in a patient with an associated head injury, as this may be interpreted as (or mask) the dilated pupil that is suggestive of an acute extradural haematoma. The lens may be damaged or dislocated and a cataract may develop. Damage to the drainage angle of the eye (which cannot be seen without a mirror contact lens and a slit lamp microscope) increases the chances of glaucoma developing in later life.

If the force of impact is transmitted to the orbit, an orbital fracture may occur (usually in the floor, which is thin and has little support). Clues to the presence of an inferior blowout fracture include diplopia, a recessed eye, defective eye movements (especially vertical), an ipsilateral nose bleed, and diminished sensation over the distribution of the infraorbital nerve. The fracture may need repair, and these

patients should be referred to an ophthalmic department.

4. What are the common patterns of penetrating injury to the eye?

Lacerations of the eyelids need specialist attention if:

- The lid margins have been torn—these must be sewn together accurately.
- The lacrimal ducts have been damaged—lacerations affecting the medial end of the eyelid are likely to damage the lacrimal ducts, and these may need to be reapposed under the operating microscope.
- There is any suspicion of a foreign body or penetrating eyelid injury—objects may easily penetrate the orbit and even the cranial cavity through the orbit.

Penetrating injuries of the eye can be easily missed because they may seal themselves and the signs of abnormality are subtle. Any history of a high velocity injury (particularly a hammer and chisel injury) should raise strong suspicions of a penetrating injury. The eye should be examined very gently without putting any pressure on the globe. Prolapse of the intraocular contents and irreversible damage can be caused if the eye and orbit are not examined carefully.

Signs to look for include a distorted pupil, cataract, prolapsed black uveal tissue on the ocular surface, and vitreous haemorrhage. The pupil should be dilated (if there is no head injury) and a thorough search made for an intraocular foreign body. If an intraocular or orbital foreign body is suspected, orbital radiographs, with the eye in up and down gaze, should be taken. If the eye is clearly perforated, a shield should be applied to protect it from any pressure and the patient should be sent immediately to the nearest eye department.

Sympathetic ophthalmia, in which chronic inflammation develops in the uninjured eye, is a potentially serious complication of any severe penetrating eye injury. The risk of this increases if a penetrating eye injury is left untreated. All penetrating eye injuries should receive immediate specialist ophthalmic management without delay.

5. Aside from definitive operating theatre procedures, briefly outline the important principles in the emergency management of each of the following:

- hyphaema
- corneal abrasion and foreign body
- ruptured globe
- intra-ocular foreign body
- retinal tears/haemorrhages/detachment
- orbital floor fracture

Hyphaema

In immediate management:

- Hyphaema should be treated as if there is an open injury — use of a protective shield, analgesia, anti-emetics and keeping the patient nil by mouth until ophthalmological review. Also, keep the patient sitting upright.
- Refer directly for an immediate ophthalmological review.

- For the specialist, the treatment objectives are to reduce the rate of re-bleeding and manage the complications of hyphaema, including glaucoma. In addition to the supportive measures outlined
- above, a cycloplegic agent (such as atropine) and a topical steroid are usually used. In rare cases, if IOP remains elevated or the patient re-bleeds, surgical intervention is required.

Corneal abrasion

- Exclude penetration or occult globe injury.
- Protect the migrating epithelial cells from toxic insults, including the regular use of anaesthetic drops.
- Eye patching can be counterproductive. Patching reduces oxygen supply and increases corneal temperature, which may promote corneal infection.
- Symptom relief can be provided with a topical cycloplegic agent. For small abrasions a stat dose of cyclopentolate 1% is appropriate. For larger abrasions use a longer acting cycloplegic, such as homatropine 2%.
- The use of topical antibiotics is controversial. Most clinicians prescribe a short course of a broad-spectrum antibiotic such as chloramycetin (Chlorsig) for 3-5 days. Topical antibiotics are toxic to the cornea, so use over a longer period in the absence of verified infection is not recommended.

Corneal foreign body

Corneal foreign bodies should be removed with a needle under topical anaesthesia; a rust ring may remain and can be removed with a small burr. Subtarsal objects can often be swept away with a cotton wool bud from the everted lid. The patient is then treated as for an abrasion. If there is any suggestion that a foreign body may have penetrated the globe the eye must be carefully examined with dilation of the pupil to allow a good view of the lens and retina. An X-ray with the eyes looking up and then the eyes looking down or a CT scan may also be indicated if an intraocular foreign body is suspected. Microsurgical techniques can be used to remove foreign bodies from the eye under direct visualisation.

Ruptured globe

Once identified no further examination of the globe should be performed but a shield should be gently placed over the eye and the patient referred for urgent ophthalmic treatment. These serious injuries, often with grave implications for sight, require careful microsurgical suturing to restore the integrity of the globe.

It is imperative that the eye is not touched or manipulated. If the object is present it should be left in situ as its removal may cause further herniation of the eye contents. Topical anaesthetic and antibiotics can be given to aid comfort and reduce the risk of infection.

Nausea or vomiting should be suppressed with the use of anti-emetics as violent head movements can aggravate the condition. Tetanus status should be confirmed and a booster dose given if required. The patient should be instructed not to eat or drink, as often surgery is required to repair the eye. It is important to advise the patient to attempt not to strain, cough, blow their nose or bend over before seeing a specialist. A pressure patch should be avoided, however a shield can be placed over the eye to provide protection. The bottom of a disposable cup is an adequate

solution if eye shields are not available. Immediate referral to a hospital and/or specialist is essential.

Vitreous or retinal pathology

An estimated 44% of patients with serious ocular injuries have vitreoretinal involvement. The most common retinal sequelae are haemorrhages, retinal detachment, retinal holes and tears, and retinal commotio (contusion). Specialist review of the fundus is required after serious ocular trauma. Patients with photopsia (flashes of light), vitreous floaters, decreased visual acuity or scotoma (visual field loss) require urgent ophthalmological review.

Orbital floor blow-out fracture

For the GP, immediate management includes:

- Protecting the eye from further injury using an eye shield.
- Providing adequate analgesia.
- Advising the patient not to blow their nose or sneeze with a closed mouth. Explosive exophthalmos can result from the passage of air from the nasal cavity into the orbital cavity.

When to refer:

All suspected orbital fractures require urgent review by an ophthalmologist and CT imaging of the orbits. Typically these patients are assessed in the emergency room. Orbital injuries that involve compression of an optic nerve require immediate decompression and close postoperative observation. Orbital injuries in which optic nerve function is preserved but there is evidence of muscle entrapment, may have surgery delayed until the swelling has subsided, usually 7-14 days after the injury.

Minimally displaced fractures without evidence of muscle or nerve entrapment can often be managed conservatively. Cosmetically unacceptable enophthalmos (sunken eye) can be a late complication of orbital fractures, and surgery may be required to bring the eye forward.

CASE TWO

Short case number: 4_21_02

Category: Nervous System & Eye

Discipline: Surgery

Setting: General practice

Topic: Acute red eye_infection

CASE



Ian is a 45-year-old sports store manager who presents with two days of swelling, irritation and redness in both eyes. The left eye is much worse than the right. He usually wears contact lenses on days that he's at work. A family member told him he has conjunctivitis and this is why he presents to see you, his GP.

QUESTIONS

1. What is the conjunctiva and where is it located?
2. What are the three common categories of conjunctivitis and how does one differentiate between these on clinical grounds?
3. Briefly outline the treatment for each type of conjunctivitis.
4. Ian uses contact lenses and this should make you particularly cautious in his assessment. Why is this so? What organisms and conditions are more likely in this group?
5. What organisms would be more likely in a neonate with infective conjunctivitis?
6. How might you decide if Ian has developed peri-orbital cellulitis in the worse eye? If this were the case, what treatment should be instituted?

Resources:

- Cronin B, Todd B, Lee G. The red eye in contact lens wearers--a high risk presentation. Aust Fam Physician; October 2007. [Accessed via Internet www.racgp.org.au].
- James B, Chew C, Bron AJ. Lecture notes on ophthalmology. Wiley-Blackwell; 2003.
- Hodge C, Lawless M. Ocular emergencies. Australian family physician. 2008;37(7):506–509.

1. A thin clear moist membrane that coats the inner surfaces of the eyelids and the outer surface of the eye. The section of the conjunctiva that coats the inner aspect of the eyelids is called the palpebral conjunctiva while that covering the outer surface of the eye is called the ocular or bulbar conjunctiva. (source: www.medicinenet.com)
2. and
- 3.

GENERAL FEATURES:

Symptoms

Patients may complain of the following:

- **Pain and irritation.** Conjunctivitis is seldom associated with anything more than mild discomfort. Pain signifies something more serious such as corneal injury or infection. This symptom helps differentiate conjunctivitis from corneal disease.
- **Redness.** In conjunctivitis the entire conjunctival surface including that covering the tarsal plates is involved. If the redness is localized to the limbus ciliary flush the following should be considered: (a) keratitis (an inflammation of the cornea); (b) uveitis; (c) acute glaucoma.
- **Discharge.** Purulent discharge suggests a bacterial conjunctivitis. Viral conjunctivitis is associated mainly with a watery discharge.
- **Visual loss.** This occurs only when the central cornea is affected. Loss of vision is thus an important symptom requiring urgent action.
- Patients with corneal disease may also complain of **photophobia**.

Signs:

The following features may be seen in conjunctival disease:

- **Papillae.** These are raised lesions on the upper tarsal conjunctiva, about 1mm in diameter with a central vascular core. They are non-specific signs of chronic inflammation. They result from fibrous septa between the conjunctiva and subconjunctiva which allow only the intervening tissue to swell with inflammatory infiltrate. Giant papillae, found in allergic eye disease, are formed by the coalescence of papillae.
- **Follicles.** These are raised, gelatinous, oval lesions about 1mm in diameter found usually in the lower tarsal conjunctiva and upper tarsal border, and occasionally at the limbus. Each follicle represents a lymphoid collection with its own germinal centre. Unlike papillae, the causes of follicles are more specific (e.g. viral and chlamydial infections).
- **Dilation** of the conjunctival vasculature (termed '**injection**').
- **Subconjunctival haemorrhage**, often bright red in colour because it is fully oxygenated by the ambient air, through the conjunctiva.

The features of corneal disease are different and include the following:

- Epithelial and stromal oedema may develop causing clouding of the cornea.

- Cellular infiltrate in the stroma causing focal granular white spots.
- Deposits of cells on the corneal endothelium (termed keratic precipitates or KPs, usually lymphocytes or macrophages).
- Chronic keratitis may stimulate new blood vessels superficially, under the epithelium (pannus) or deeper in the stroma. Stromal oedema, which causes swelling and separates the collagen lamellae, facilitates vessel invasion.
- Epithelial erosions are punctate or more extensive patches of epithelial loss, which are best detected using fluorescein dye and viewed with a blue light.

BACTERIAL CONJUNCTIVITIS

Patients present with:

- redness of the eye;
- discharge;
- ocular irritation.

The commonest causative organisms are *Staphylococcus*, *Streptococcus*, *Pneumococcus* and *Haemophilus*. The condition is usually self-limiting although a broad-spectrum antibiotic eye drop will hasten resolution. Conjunctival swabs for culture are indicated if the condition fails to resolve. Ophthalmia neonatorum, which refers to any conjunctivitis that occurs in the first 28 days of neonatal life, is a notifiable disease. Swabs for culture are mandatory. It is also important that the cornea is examined to exclude any ulceration.

The commonest organisms are:

- Bacterial conjunctivitis (usually Gram positive).
- *Neisseria gonorrhoea*. In severe cases this can cause corneal perforation. Penicillin given topically and systemically is used to treat the local and systemic disease respectively.
- Herpes simplex, which can cause corneal scarring. Topical antivirals are used to treat the condition.
- Chlamydia. This may be responsible for a chronic conjunctivitis and cause sight-threatening corneal scarring. Topical tetracycline ointment and systemic erythromycin is used to treat the local and systemic disease respectively.

VIRAL CONJUNCTIVITIS

This is distinguished from bacterial conjunctivitis by:

- a watery and limited purulent discharge;
- the presence of conjunctival follicles and enlarged pre-auricular lymph nodes;
- there may also be lid oedema and excessive lacrimation.

The conjunctivitis is self-limiting but highly contagious. The commonest causative agent is adenovirus and to a lesser extent Coxsackie and picornavirus. Adenoviruses can also cause a conjunctivitis associated with the formation of a pseudomembrane across the conjunctiva. Certain adenovirus serotypes also cause a troublesome punctate keratitis. Treatment for the conjunctivitis is unnecessary unless there is a secondary bacterial infection. Patients must be given hygiene instruction to minimise the spread of infection (e.g. using separate towels). Treatment of keratitis

is controversial. The use of topical steroids damps down symptoms and causes corneal opacities to resolve but rebound inflammation is common when the steroid is stopped.

ALLERGIC CONJUNCTIVITIS

This may be divided into acute and chronic forms:

1. Acute (hayfever conjunctivitis). This is an acute IgE-mediated reaction to airborne allergens (usually pollens). Symptoms and signs include:

- (a) itchiness;
- (b) conjunctival injection and swelling (chemosis);
- (c) lacrimation.

2. Vernal conjunctivitis (spring catarrh) is also mediated by IgE. It often affects male children with a history of atopy. It may be present all year long. Symptoms and signs include:

- (a) itchiness;
- (b) photophobia;
- (c) lacrimation;
- (d) papillary conjunctivitis on the upper tarsal plate (papillae may coalesce to form giant cobblestones);
- (e) limbal follicles and white spots;
- (f) punctate lesions on the corneal epithelium;
- (g) an opaque, oval plaque which in severe disease replaces an upper zone of the corneal epithelium.

Initial therapy is with antihistamines and mast cell stabilisers (e.g. sodium cromoglycate; nedocromil; lodoxamide). Topical steroids are required in severe cases but long-term use is avoided if possible because of the possibility of steroid induced glaucoma or cataract.

Contact lens wearers may develop an allergic reaction to their lenses or to lens cleaning materials leading to a *giant papillary conjunctivitis (GPC)* with a mucoid discharge. Whilst this may respond to topical treatment with mast cell stabilisers it is often necessary to stop lens wear for a period or even permanently. Some patients are unable to continue contact lens wear due to recurrence of the symptoms.

4.

Contact lens associated keratitis is a severe, sight threatening condition. Pathogens such as *acanthamoeba* and staphylococcal species can be responsible, however, the majority of cases involve *Pseudomonas aeruginosa*. The intense inflammatory reaction can lead to a hypopyon and endophthalmitis (infection in the vitreous), potentially leading to enucleation or blindness.

Not only do the severe consequences of contact lens associated keratitis (Figure 2) make it an important subgroup of patients to recognise, but the common treatment of suspected eye infections, chloramphenicol, is not effective against *pseudomonas* (or many other common contact lens associated pathogens).^{3,5} Therefore, the common safety net management plan will not only allow *pseudomonas* to continue growing, but wastes valuable time in delaying appropriate treatment.

Once a patient is recognised as being at risk of contact lens keratitis, management should include:

- immediate removal of the lenses
- ensuring the patient keeps their lenses and lens case so that they can be sent for culture
- staining of the cornea with fluorescein and full examination including visual acuity (remember, not all cases of bacterial keratitis will stain with fluorescein)
- if an epithelial defect or obvious infiltration (opaque zone) is present, then the patient should be referred immediately to an ophthalmologist for assessment the same day. If no obvious ulcer is present then the patient should still be referred for urgent ophthalmic assessment. In the rural setting, phone advice from an ophthalmologist should be sought before deciding on a management plan.
- Definitive ophthalmic management will generally include scraping the corneal epithelium and stroma and directly plating this to culture media to obtain isolates and sensitivities. Severe cases will involve admission for round-the-clock broad-spectrum topical fortified antibiotics, pending sensitivity results. In some cases, systemic antibiotics may also be used.

Summary

In the general practice setting, a few simple maxims will ensure safe and simple practice for the general or emergency department practitioner:

- a painful eye in a contact lens wearer should be considered to be sight threatening bacterial keratitis until proven otherwise, and
- the most likely pathogen in contact lens associated bacterial keratitis is *Pseudomonas aeruginosa* which is not sensitive to chloramphenicol.

5. Neonatal conjunctivitis

Extra consideration must be given to chlamydia, gonorrhea and Herpes simplex.

6. Orbital cellulitis

Orbital cellulitis is a rare but potentially life threatening condition that exhibits both ocular and general symptoms. The typical patient is younger, with general malaise and rapid onset of fever. Commonly the eye will be proptosed and the patient will describe severe pain on ocular movements. The eye and surrounding tissue will be chemotic and visual acuity may be poor.

History may indicate a specific cause, including extension of an existing infection of the paranasal sinuses or adjacent structures and from inoculation as a result of recent facial trauma or surgery. It is important to differentiate orbital cellulitis from the more common preseptal cellulitis.

Preseptal cellulitis rarely exhibits localised swelling which may mimic proptosis, however the eye is not proptosed and ocular movements are normal. Preseptal cellulitis rarely affects vision or presents with red eye. Untreated orbital cellulitis can lead to meningitis, brain abscess and cavernous sinus thrombosis, therefore immediate specialised care and hospitalisation is required. Signs of deterioration may include binocular involvement and an increase in neurological symptoms.

CASE THREE

Short case number: 4_21_03

Category: Nervous System & Eye

Discipline: Surgery

Setting: General Practice

Topic: Acute red eye_inflammation

CASE



Sally is 39 years old and presents to your general practice complaining of a red & sore right eye. It has been this way for around 48hrs and is getting worse. In addition, her vision from that eye seems a little blurred and there is a watery discharge. There is no history suggestive of a foreign body

Sally's background health includes asthma, chronic lower back pain and genital herpes. She has tried some over the counter lubricant drops which only helped a little.

QUESTIONS

1. In assessing the unilateral red eye, what common or serious disorders should be considered
2. How would you decide whether Herpes simplex might be the cause of Sally's presentation?
3. Some old test results in her file, suggest that Sally is positive for the HLAB27 marker. What ocular diagnosis is commonly associated with this result and what historical and examination findings would be consistent with such a diagnosis?
4. Aside from HLAB27 antigen status, what are some other causes for this diagnosis?
5. Sally's ocular problem is indeed caused by the diagnosis that you suspected in Q3. What treatment can be offered and what is the prognosis?

Resources:

- Hodge C, Dunlop A. Unilateral red eye. Eye series-16. Australian family physician. 2004;33(6):443.
- James B, Chew C, Bron AJ. Lecture notes on ophthalmology. Wiley-Blackwell; 2003.
- Sharma NS, Ooi JL, Li MZ. A painful red eye. Australian family physician. 2009;38(10):805.

ANSWERS

1.

Table 1. Differential diagnosis of a unilateral red eye

Condition		Features
Conjunctivitis	Allergic	• Usually bilateral • Itch • History of atopy
	Viral	• Symptoms often start in one eye and then develop in the other a few days later • Watery discharge • Pre-auricular lymphadenopathy • ± recent viral illness • Consider herpes simplex virus if there is history of cold sores and/or a dendritic corneal lesion on examination
	Bacterial	• Purulent discharge; often bilateral • Consider Pseudomonas/Acanthamoeba keratitis in contact lens wearers
Corneal foreign body		• Irritation • ± ocular trauma/metal work
Acute anterior uveitis		• Pain • Photophobia • Blurred vision • ± irregular pupil (posterior synechiae) • Slitlamp examination – cells in anterior chamber
Acute angle closure (acute glaucoma)		• Severe pain and headache • Photophobia • Visual impairment • Nausea, vomiting • Pupil fixed and mid-dilated • Cloudy cornea
Microbial keratitis		• Suspect (especially in contact lens wearer) • May get infiltrate or abscess (white lesion on cornea) • Needs prompt specialist assessment
Endophthalmitis		• History of previous intraocular surgery • Severe pain is typical • Hypopyon (pus in anterior chamber) is often present • If endogenous, patient may be immunocompromised • Needs urgent specialist treatment (within hours not days)

The presentation of a unilateral, painful red eye can indicate the presence of several conditions. These may include bacterial conjunctivitis, corneal or intraocular inflammation, ocular foreign body and acute angle closure glaucoma. Although commonly bilateral in presentation, bacterial conjunctivitis may present in one eye. The patient will generally complain of a red, irritable eye. Purulent discharge is the hallmark of a bacterial infection. The patient may complain of photophobia if there is associated punctate epitheliopathy or peripheral corneal infiltrates.

Vision can also be affected in severe cases. Keratoconjunctivitis sicca (dry eyes) is a relatively common condition. On presentation the patient will complain of grittiness, excess tearing, photophobia and red eyes, with symptoms worse later in the day, or in hot or dry conditions. This condition rarely significantly threatens sight.

An ocular foreign body will cause localised irritation and inflammation. Patients will commonly produce excessive tearing as a result of the irritation. Similarly, the patient generally complains of light sensitivity. The vision may or may not be affected. Acute angle closure glaucoma is characterised by a sudden and severe increase in intraocular pressure. On presentation the patient will complain of pain, extreme photophobia and sudden deterioration of vision. The eye will be injected around the limbus. The pupil will be semi dilated and often unreactive to light. Vision is affected by the presence of severe corneal oedema.

Uveitis is an intraocular inflammation that involves the uveal tract (iris, ciliary body and choroid). Uveitis can be further classified depending on which structures are affected. Anterior uveitis, involving the iris and occasionally also the anterior part of the ciliary body, will develop over several days beginning with general sensitivity and progressing to photophobia, pain, redness over the ciliary body (Figure 1) and visual loss. Keratic precipitates (KP), small cellular deposits from the anterior chamber on the posterior surface of the cornea, can be seen on close examination.

2.

HERPES SIMPLEX KERATITIS Type 1 herpes simplex (HSV) is a common and important cause of ocular disease. Type 2 which causes genital disease may occasionally cause keratitis and infantile chorio-retinitis. Primary infection by HSV1 is usually acquired early in life by close contact such as kissing.

It is accompanied by:

- fever;
- vesicular lid lesions;
- follicular conjunctivitis;
- pre-auricular lymphadenopathy;
- most are asymptomatic.

The cornea may not be involved although punctate epithelial damage may be seen. Recurrent infection results from activation of the virus lying latent in the trigeminal ganglion of the fifth

cranial nerve. There may be no past clinical history. The virus travels in the nerve to the eye. This often occurs if the patient is debilitated (e.g. psychiatric disease, systemic illness, immunosuppression).

It is characterised by the appearance of dendritic ulcers on the cornea. These usually heal without a scar. If the stroma is also involved oedema develops causing a loss of corneal transparency. Involvement of the stroma may lead to permanent scarring. If corneal scarring is severe a corneal graft may be required to restore vision. Uveitis and glaucoma may accompany the disease. Disciform keratitis is an immunogenic reaction to herpes antigen in the stroma and presents as stromal clouding without ulceration, often associated with iritis.

3. Uveitis. (Acute anterior uveitis).

The patient may complain of:

- ocular pain (less frequent with posterior uveitis or choroiditis);
- photophobia;
- blurring of vision;
- redness of the eye.
-

(Posterior uveitis may not be painful).

On examination:

- The visual acuity may be reduced.
- The eye will be inflamed in acute anterior disease, mostly around the limbus (*ciliary injection*). Inflammatory cells may be visible clumped together on the endothelium of the cornea particularly inferiorly (*keratitic precipitates or KPs*).
- Slit lamp examination will reveal aqueous cells and flare. If the inflammation is severe there may be sufficient white cells to collect as a mass inferiorly (*hypopyon*).
- The vessels on the iris may be dilated.
- The iris may adhere to the lens (*posterior synechiae or PS*).
- The intraocular pressure may be elevated.
- There may be cells in the vitreous.
- There may be retinal or choroidal foci of inflammation.
- Macular oedema may be present.

4.

CAUSES OF UVEITIS		
Infectious	Associated with systemic disease	Ocular disease
Toxoplasmosis	Ankylosing spondylosis	Advanced cataract
Postoperative infection	Sarcoidosis	Sympathetic ophthalmitis
Fungal	Reiter's disease	Retinal detachment
CMV	Behçet's disease	Angle closure glaucoma
Herpetic	Psoriatic arthritis	Intraocular tumours
Tuberculosis	Juvenile chronic arthritis	
Syphilis	Inflammatory bowel disease	
Metastatic infection		
Toxocara		

Table 9.1 Table showing some causes of uveitis (this is not an exclusive list).

5. The aims of ocular therapy are to resolve the inflammation, relieve symptoms, return normal vision and prevent ocular scarring. Although the course of an acute uveitis attack is generally 2–6 weeks, referral to an ophthalmologist within 24 hours is essential to an optimum outcome.

Corticosteroid drops such as prednisolone are the preferred initial therapy. Due to the possible side effects of steroid drops it is important the patient is carefully screened before and during treatment. Possible short-term complications include acute intraocular pressure rise, which can quickly lead to severe visual loss. Uveal inflammation may lead to ciliary spasm and cause periocular pain. Furthermore, the uveitis patient is at increased risk of posterior synechiae. Long acting cycloplegics such as homatropine and atropine, work to relieve the pain and reduce the possibility of adhesions. If inflammation persists, periocular or oral steroids may be required. Treatment is tapered over 6 weeks despite symptomatic and clinical quiescence to avoid bounce back inflammation.

Treatment of the systemic findings is dependent on symptoms. NSAIDs may be used to treat sore or painful joints. If inflammation persists, oral or injectable corticosteroids or these medications. Consultation with a rheumatologist is required. Physical therapy to improve flexibility and maintain correct posture is considered a useful adjunct to medical treatment in cases of ankylosing spondylitis.

CASE FOUR

Short case number: 4_21_04

Category: Nervous System & Eye

Discipline: Surgery

Setting: Emergency department

Topic: Chemical burns of the eye

CASE



Working in casualty late one night, the switchboard puts through a call from Alfonse, a 43 year old cleaning contractor.

He has been working at an industrial plant and has accidentally splashed some strong cleaning fluids into his eyes.

QUESTIONS

1. What would you want to find out from Alfonse and what instructions would you give to him?
2. Alfonse follows your instructions and, after some time, presents in person (brought in by a family member) to your department. Realising that irrigation must take precedence over examination, you arrange for this to continue. What liquid or solution should be used?
3. What indicator might tell you that it is safe to cease irrigation and perform a thorough examination?
4. Are there any adjuncts that might make examination easier?
5. After phoning the state poison's hotline, you learn that the cleaning fluid is a potent alkali. You provide appropriate emergency care to Alfonse and then decide to phone the ophthalmologist on-call. What examination findings might you relay to the specialist, so that he or she might get a clear sense of the severity of the injury?
6. Beyond first aid, what other measures or procedures are sometimes considered in the further treatment of severe chemical injuries to the eyes?

Resources:

- Hodge C, Lawless M. Ocular emergencies. Australian family physician. 2008;37(7):506–509.
- Rudkin AK, Lake S. How to treat: Eye Injuries. Australian Doctor [Internet]. 2009 May 22; Available from: http://www.australiandoctor.com.au/http/pdf/AD_027_034_MAY22_09.pdf

ANSWERS

1. It is important to determine the nature of the fluid. A brand name can be useful for making further enquiries as to the ingredients or likelihood of significant injury.

Regardless of the nature of the chemical injury, immediate, copious irrigation of the eye is essential to limit the damage to the eye. The relative pH level of the eye increases more rapidly and is ultimately higher when irrigation is commenced later. Significant delay will affect the final prognosis.

If the patient has telephoned, they should be instructed to irrigate the eye for at least 20–30 minutes before arriving.

2. Ideally an eye wash or buffering solution should be used to help balance the pH level of the eye. Tap water is not as effective, although it remains an effective rinsing solution and is readily available.
3. A return to a normal pH is critical before ceasing irrigation. Removal of residual chemicals and materials is critical. pH might be gauged with litmus paper or with a urinalysis dipstick.
4. Instilling local anaesthetic will make examination more comfortable.
5. Following mild chemical burns the eye will be hyperaemic and the conjunctiva chemosed or swollen. The cornea may have a hazy appearance. In severe cases the eye will appear white due to the relative ischaemia of the limbal and conjunctival vessels. The iris and pupil will also be difficult to see due to the complete opacification of the cornea.

Table 2. Roper-Hall classification of the severity of ocular burns

Grade	Prognosis	Cornea	Conjunctiva/limbus	Referral
I	Good	Epithelial damage	No limbal ischaemia	Within 24 hours
II	Good	Corneal haze	<1/3 limbal ischaemia Iris details visible	Immediate
III	Guarded	Total epithelial loss	1/3–1/2 limbal ischaemia Stromal haze Iris details obscured	Immediate
IV	Poor	Cornea opaque	>1/2 limbal ischaemia Iris and pupil obscured	Immediate

6.

Any patient with a corneal injury from a chemical injury should be referred (after irrigation of the eye) to an emergency ophthalmology clinic.

The aim of further management is to promote epithelialisation and inhibit the inflammatory response. Mild injuries (grade 1 or 2) are treated with a short course of steroids, cycloplegics and prophylactic antibiotics. In more severe injuries, systemic and topical ascorbic acid, topical sodium citrate and systemic tetracyclines are given.

In severe injuries, surgery has a role, including limbal stem cell transplantation, human amniotic membrane transplantation and other procedures aiming to revascularise the limbus and restore the limbal cell population.

CASE FIVE

Short case number: 4_21_05

Category: Nervous System & Eye

Discipline: Surgery

Setting: General practice

Topic: Sudden loss of vision

CASE

You are working as a general practitioner. Shannon is 38 years old patient who appears quite worried. She is having severe trouble seeing. She thinks it might have been brought on by the stress of her job in HR. The visual disturbance came on very quickly about forty minutes ago. It seems to be mainly on the left side.

QUESTIONS

1. What might you ask Shannon to determine if this is a problem of one eye (or associated optic nerve), versus a problem of the visual pathway posterior to these structures?
2. What are the common causes of sudden visual loss anterior to the optic chiasm? How might a thorough history make these diagnoses more or less likely? (Consider especially speed of onset, age, gender, and associated symptoms/illness/risk factors).
3. Shannon's visual loss is confined to the left eye only. There is no headache and the problem doesn't clear with blinking. She can just manage to count fingers with the left eye, but can't read the top letter of a Snellen chart. There is reduced direct pupillary constriction apparent in the left eye. What sorts of signs might be sought through fundoscopy?
4. Aside from fundoscopy, what other kinds of examination should be performed?
5. Fundoscopy reveals extensive retinal haemorrhaging, tortuous veins and a swollen optic disc (all in the left eye; right side normal). What is the diagnosis? Are further investigations needed?
6. What treatment might be recommended for Shannon and what is her prognosis?

Resources:

- Goold L, Durkin S, Crompton J. Sudden loss of vision: history and examination. Australian family physician. 2009;38(10):764–768.
- James B, Chew C, Bron AJ. Lecture notes on ophthalmology. Wiley-Blackwell; 2003.

1. It is critical to determine whether the visual loss involves one eye or both. The patient must be specifically questioned as some may, for example, confuse a left homonymous field loss for a loss of vision in the left eye. Bilateral visual disturbance rarely indicates bilateral ocular or optic nerve disease, but more commonly reflects disease involving or posterior to the optic chiasm, such as pathology of the optic tract or cerebral cortex, including impaired posterior cerebral circulation or migraine.

2. Duration of vision loss

The duration of the visual loss, the degree to which it recovers, and any associated symptoms are also important. Visual loss lasting for seconds to minutes may be due to embolic phenomena or optic nerve head drusen – deposits of extracellular material under the retina; whereas visual loss occurring over days to weeks may indicate either a compressive, infiltrative or inflammatory pathology. A visual obscuration that recovers with blinking is likely to be due to tear film abnormalities, whereas one that occurs in the setting of headache, photophobia and phonophobia, may be due to migraine.

Associated symptoms

It is also important to elicit associated symptoms such as double vision, sore eyes, red eyes, haloes and flashing lights. These symptoms may be associated with other cranial nerve lesions, ocular surface disorders, angle closure glaucoma and migraine. Past history of migraine, childhood motion sickness, family history of migraine, transient ischaemic attack or stroke, and atherosclerotic risk factors including smoking or recent trauma (e.g. neck manipulation, motor vehicle accident, roller coaster ride) may also provide clues to the aetiology. Acute angle closure glaucoma is typified by vision loss associated with intense pain, haloes, conjunctival injection, nausea, vomiting and photophobia.

Table 1. Causes of sudden vision loss

Onset	Duration	Differential diagnosis	Associations
Seconds	Seconds	Tear film abnormalities	Clears with blinking
		Papilloedema, optic disc drusen, vitreous debris (floaters)	Associated with eye movement
Seconds	Variable	Orbital tumour	Associated with other orbital cranial nerve palsies
Seconds	Less than 10 minutes	Carotid occlusive disease, vertebrobasilar disease	Cardiovascular risk factors
		Cardiac emboli	Cardiovascular risk factors
		Giant cell arteritis	Headache, jaw claudication, scalp tenderness, diplopia, shoulder girdle pain, weight loss, night sweats
		Postural hypotension	Dehydration, cardiovascular disease, presyncope
Minutes	Minutes to hours	Transient angle closure glaucoma	Pain, haloes, blurred vision, red eye
		Migraine	Zig-zag' lights, headache
Seconds	Minutes to indefinite	Retinal artery occlusion	Cardiovascular risk factors
		Retinal vein occlusion	Cardiovascular risk factors
		Retinal detachment	Flashing lights, floating spots, 'curtain' progressing from side of visual field
		Optic neuritis	Pain, exacerbated by exercise or elevated body temperature (Uhthoff phenomenon)

3.

Fundoscopy

Fundoscopy should be performed following dilation if the pupil examination is normal and trauma or acute angle closure glaucoma is not suspected. Phenylephrine 2.5% and tropicamide 1% are often used with a duration of action of 2–6 hours. Caution should be exercised in elderly or hypertensive patients when considering phenylephrine use. Before examining retinal and optic nerve features, the red reflex should be assessed in each eye. This may be obscured by media opacities such as cataract or vitreous haemorrhage, or intraocular pathologies such as retinal detachment. As a general rule of thumb, the quality of the red reflex often gives an indication of the limitation of visual impairment due to media opacities.

The optic disc may be located by following blood vessels and should be inspected for pallor, swelling, haemorrhage or gross cupping. Retinal blood vessels may reveal signs of hypertensive retinopathy, emboli, or venous congestion. The macular, found temporal to the optic disc, should be examined last. Particularly in the undilated eye, this minimises patient discomfort and reduces the effect of pupillary constriction on the remainder of the fundus examination. A cherry-red spot is indicative of retinal ischaemia (as the macula supplied by the choroidal vascular bed beneath becomes more prominent relative to the surrounding ischaemic retina supplied by the retinal vessels).

4. Careful examination of the cranial nerves may determine the cause of vision loss through reconciliation of symptoms with anatomical structures. Vision loss in concurrence with lesions of the intraorbital **cranial nerves** (optic, oculomotor, trochlear, the ophthalmic branch of the trigeminal and abducens) may indicate an orbital mass, while central retinal vein occlusion and

involvement of the ophthalmic and maxillary branch of the trigeminal nerve may suggest cavernous sinus thrombosis.

Historical features and examination findings may also lead the physician to check **blood pressure** or **blood sugar** levels and to **auscultate the carotid arteries** for bruits or the heart for murmurs. It may also be necessary to check the function of more inferior cranial nerves, or to conduct a more thorough neurological examination, especially in the setting of a homonymous field defect. Where GCA is suspected it is also imperative to **palpate the temporal arteries** looking for tenderness or reduced pulsatility.

5. This is characteristic of a central retinal vein occlusion. Investigation of a CRVO includes vascular and haematological work-up to exclude increased blood viscosity. Central retinal vein occlusion is also associated with raised ocular pressure, diabetes and hypertension.

6. TREATMENT

Retinal laser treatment is given if the retina is ischaemic to prevent the development of retinal and iris new vessels. Laser treatment may improve vision in some patients with a branch retinal vein occlusion by reducing macular oedema.

PROGNOSIS

The vision is usually severely affected in central, and often in branch, vein occlusion and usually does not improve. Younger patients may fare better, and there may well be some visual improvement.